

Course: ML2302-Modelling, Simulation and
Optimization of Sustainable Production

Project Work: SSAB Case-Arena B1

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1. Intro

Company and case background

The Swedish steel manufacturing company SSAB is active worldwide with the main operations in Sweden, Finland and the US. The total workforce consists of around 15,000 employees, whereas 2,500 of these are in Oxelösund (*Historia*, 2024). The current constellation of the company started in 1978, but the earliest operations began already in the late 1800's (SSAB, 2023). Their products today include a variety of high quality steel and in this case study the specific product is their HARDOX and STRENX steel plates.

SSAB is developing a new manufacturing technique, which requires updates to their current process flow and facility restructuring at their Oxelösund site. They investigate how to increase their quenching capability, and how it would need facility restructuring on their site in Oxelösund. As quenching is a vital part of steel production with rapid actions of heating and cooling to get desired properties of the material the optimization is essential (Sild, 2023). Our goal is to maximize the utilization of their furnace. The steel requires high temperatures and heavy equipment to be handled, which is not to be solved by pure trial and error in real life, due to high cost and unnecessary equipment wear. Instead it will be simulated to approximate how the future production will perform and be executed. By optimizing the utilization of the furnace through a steady sequence input of plates SSAB aims to reduce production wastes. This would be noticed in different tracked KPIs such as lead time and energy use. The simulation will be built upon the logistic process of moving, storing and retrieving the items with the consideration of the different conditions for the items, machines in use and pile dimensions based on historical data provided by SSAB.

The current state

The current operations of the Oxelösund facility reports production volumes of **159,501,042 kg** during February 1, 2024 – May 29, 2024 within the **steel plate production process, covering 11 production groups**. The most common size of the product is **4.0 mm** on about **5.57% of the products**, which consist of **5.56%** of the output volume.

Research questions and approach

The specific prioritized questions to be answered in this case are as following:

1. How many piles and/or overhead cranes do we need to have to be able to process 7500 tons per week?
2. How much idle running will we have in the furnace?
3. How do we maximize the utilization of the furnace based on a given schedule?

To address research questions the simulation software Arena will be used. Benefits by using this program consist of detailed scenario analysis and access to adjust parameters between scenario-comparison. This case study will be concise of different parts. The Methodology will present how this simulation has been created, the theory behind it and the validation/verification process. The Results section will demonstrate how the model has been performing, and what had to be adjusted. Then Findings presents the extracted information of what this optimized flow would achieve. Finally the report is being concluded with Discussion and Conclusion interpreting the simulation, how this can be applied in the real world and what could be improved in future simulation.

2. Method

2.1 Overview

This section of the report covers the methodology followed throughout the modelling, simulation and optimization processes of the quenching facility at SSAB, Oxelösund.

2.2 Conceptualization and Model Development

Conceptualization is a key step in simulation modeling, simplifying real-world processes into structured representations. This stage ensures a clear understanding of the system's workflow, resources, and constraints before implementing the simulation model in Arena. To achieve this, the Product-Process-Resource (PPR) model and IDEF0 conceptual model were used. These frameworks are commonly applied in process planning and resource allocation, as described by (Fechter and Seeber, 2018).

The PPR model provides an initial high-level view of system components, while IDEF0 helps break down detailed process interactions, making them useful for modeling industrial workflows.

2.2.1 Product-Process-Resource (PPR) Model

The PPR model serves as the starting point for structuring the system, categorizing it into three main components accordingly for our case of SSAB:

- **Product:** The steel plates being processed.
- **Process:** The sequence of operations, including arrival, sorting, transport, and quenching.
- **Resources:** The physical elements required to perform the process, such as conveyor belts, cranes, and the furnace.

By organizing the system this way, the PPR model (Fig.1.) provided a structured foundation for analyzing the interactions between products, processes, and resources. This initial step was critical in ensuring that all key elements of our case study were considered before moving into more detailed modeling.

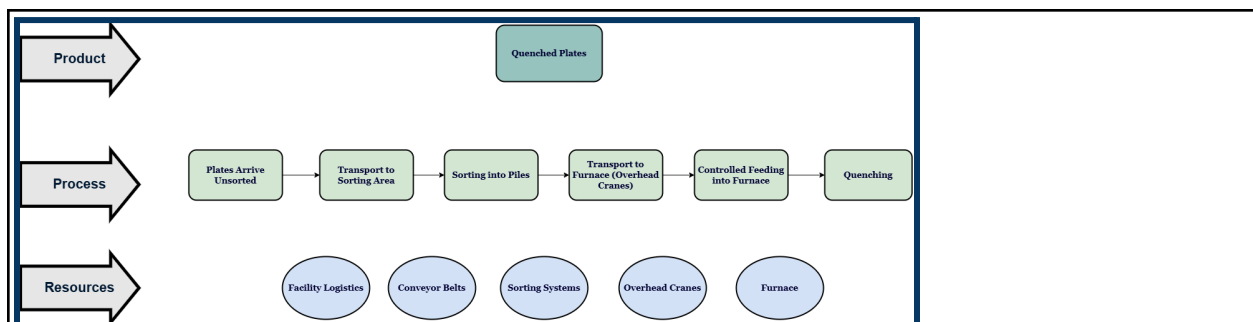


Fig.1. PPR-model

2.2.2 IDEF0 Conceptual Model: Initial Version

Building upon the PPR model, the first iteration of the IDEF0 conceptual model (Fig.2.) was developed to visualize the quenching process in more detail. The initial version outlined the major steps:

1. **Arrival of plates** at the facility.
2. **Transport to sorting areas** via conveyor belts.
3. **Sorting of plates** based on thickness.
4. **Transporting from one conveyor belt to another** based on the maximum weight capacity
5. **Transport to the quenching system** using overhead cranes.
6. **Quenching process** where plates are heat-treated.

At this stage, assumptions were made based on available process descriptions and facility documentation. However, further refinements were needed to improve accuracy.

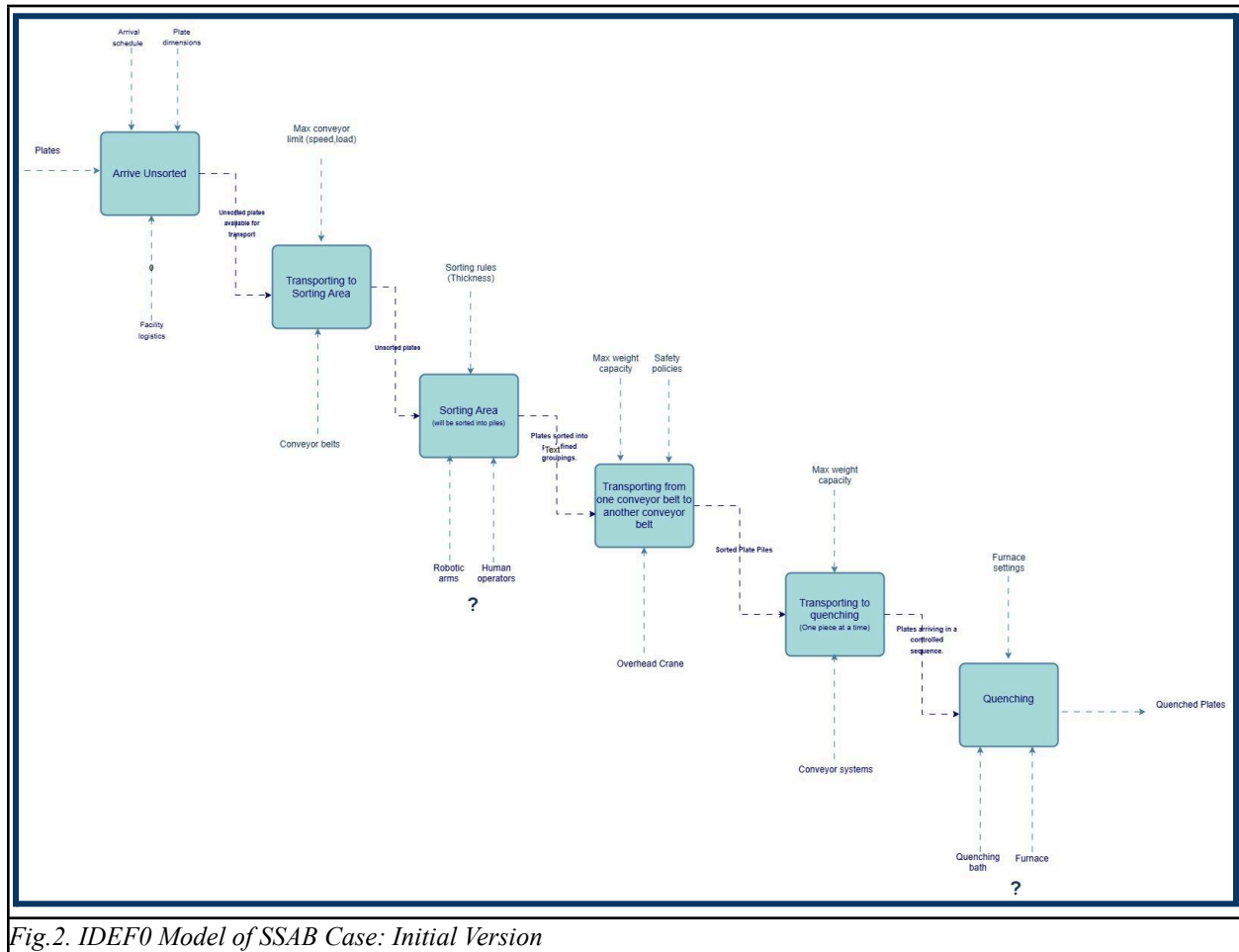


Fig.2. IDEF0 Model of SSAB Case: Initial Version

2.2.3 Refining the Model: Updates After Site Visit

The site visit to SSAB provided a clearer understanding of the facility's operations, leading to several refinements in the conceptual model (Fig.3.). These updates ensure that the simulation reflects actual processes more accurately.

1. **Bay Assignment in Sorting:** Initially, sorting was based only on thickness. The revised model incorporates bay assignment rules to improve material flow.
2. **Handling of Non-Conforming Plates:** A disposal step was introduced to account for plates that do not meet processing requirements.
3. **Transport to Furnace:** The original model assumed plates would go directly to quenching after sorting. However, it was clarified that plates must first be transported to the furnace before reaching the quenching stage.
4. **Controlled Quenching Sequence:** The updated model explicitly includes the role of furnace speed in regulating the sequence of quenching.
5. **Control Room Operations:** Human operators responsible for monitoring sorting and furnace feeding were added to reflect real operational practices.

These modifications help align the model with SSAB's actual workflow, making the simulation a more reliable representation of the system.

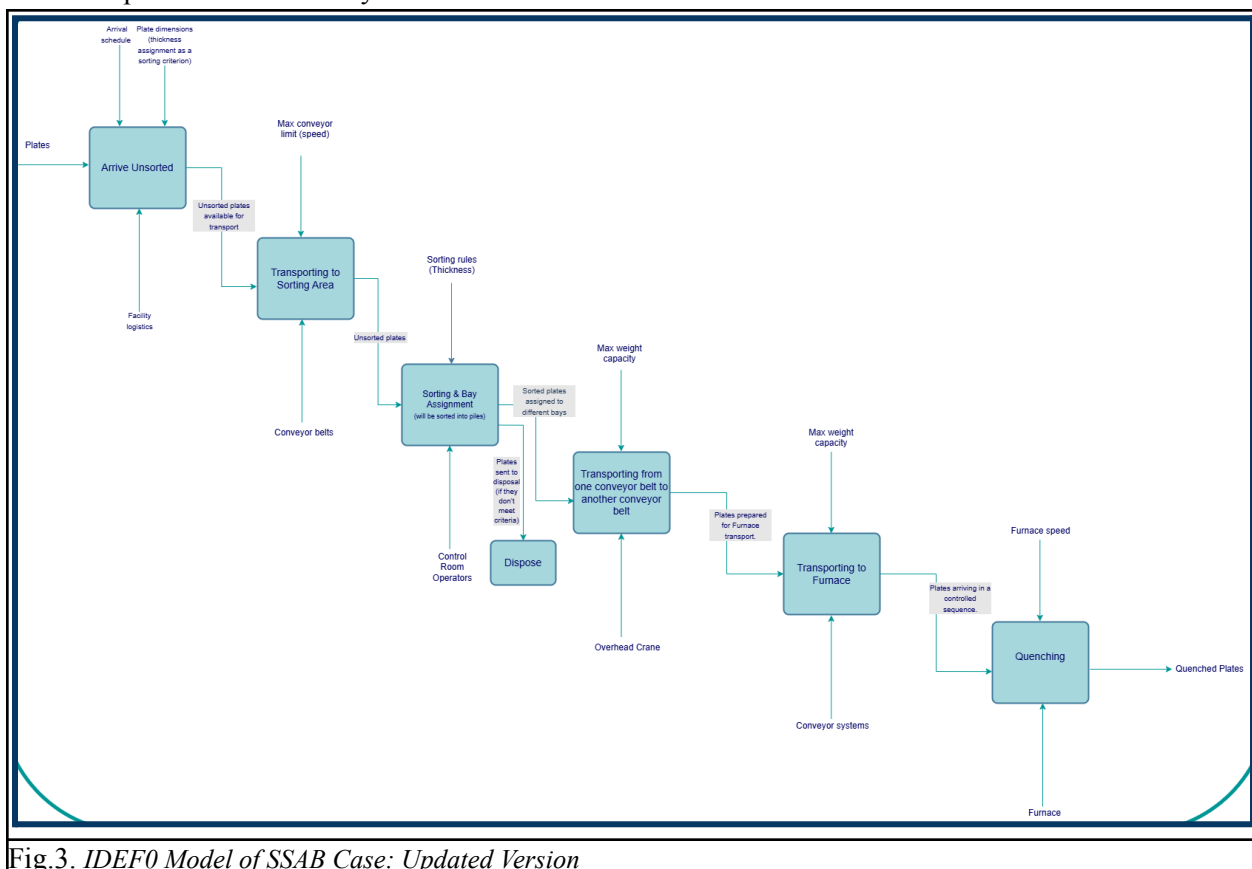


Fig.3. IDEF0 Model of SSAB Case: Updated Version

Model Implementation on Arena

The Arena model was developed to simulate the material flow and resource interactions within the quenching facility, focusing on key processes such as sorting, transport, and process operations. Initially, the model was built in a simplified form, an experimental design, capturing the core system behavior without unnecessary complexity. It started by generating entities based on probability distributions, assigning attributes to the entities, and routing them through various processes. Resources like cranes, conveyors, and the furnace were defined with capacity constraints to reflect real-world operations, while queues were implemented to manage delays and optimize efficiency. As the model developed, additional

details were introduced, including refined resource constraints, queue logic, and statistical distributions to improve accuracy. Instead of striving for an exact replica, which would be near impossible, the model was adjusted to ensure it represented key system dynamics effectively. This helps balance computational cost and reality while providing valuable insights into system performance.

Assumptions

Simplifications are discussed throughout the report, below are the assumptions that have led us to take simplification decisions in creating the model.

1. Assumed that different thickness groups do not require gaps as they enter the furnace – the temperature can be increased or decreased within the 90m length of the furnace to match the requirements of the incoming plate. Decided after the discussions in site visit on 14th February.
2. Up & down movement of the crane is always the same time, represented by a delay in the model; so the current height of the pile does not affect the time of crane movement. E.g. if a pile is empty the crane would have to travel 1.8 meters more than if it is almost full.
3. Each bay fits 7 piles between the two conveyor belts, rest of the piles are outside the second conveyor belt.
4. The width range in piling rules assumed to fit all the plates created – in reality some plates were outliers but the number of plates were small, so could be ignored.
5. Centering device was implemented once at the start of the bay entry, assuming that the plates would always remain centered if there were not outside forces as the cranes move plates from conveyor to pile to conveyor. 1 crane per bay, all piles assigned in a bay request and use the same crane.

2.3 Data Acquisition

According to (Banks, 2010) input data is the driving force of the simulation. The distribution of arrival time and the resources required make up this data. The steps required can be summarized in four steps; Collect Data, Identify Distribution, Choose Parameters, Evaluate for goodness of fit. In this case, the data was collected and provided to us from past operational records, therefore only steps 2-4 were performed. Similarly the equipment (resource) specifications and past records were also provided for our use.

To reflect the real world, processing constraints such as maximum pile height, crane loading restrictions etc. were set. Similarly, resource availability was also covered such as the number of conveyor belts, the requirement for one-by-one quenching, and the range of dimensions of plates suitable for quenching. This resulted in the understanding of data and cleaning based on the dimension constraints where historical data was no longer appropriate for the new facility. Where data was limited or unavailable, SSAB was contacted for assistance.

Distributions and Statistical Analysis

Observing and analyzing data is important for determining whether it is adequate for selecting the appropriate distribution. Given that we were provided with data from an old furnace plant, this data had to be interpreted to reflect real-world system behavior. To ensure accuracy, the best-fitting probability distribution for key process parameters was identified using goodness-of-fit tests, probability plots, and statistical analysis. There are different ways of choosing the data distribution. A popular method is for histograms to be created in both to infer a known pdf (probability density function) which is used to choose the best family of distribution. In this case, Arena Input Analyzer was used to identify

distributions, providing the square error to show how close of a fit it is to each distribution, where the closest one could be selected. It calculates a p-value to measure how well a distribution fits the data. A larger p-value suggests a better fit, but software tools might pick flexible distributions like Gamma or Weibull over simpler ones like Exponential, since they match the data more closely. However, a closer match might not suggest the distribution is the right choice for the model. Since p-values don't show where the lack of fit occurs, visual checks like histograms, P-P plots, and Q-Q plots are needed to confirm the fit. The final choice would be based on both statistical results and knowledge of the system, rather than just taking the highest p-value. The tests used by Arena consist of the two following:

Chi-squared test

It checks to see if a dataset follows a specific probability distribution by comparing the observed frequencies in data bins to expected frequencies from a theoretical distribution. If the difference between observed and expected values is too large, the null hypothesis (that the data fits the chosen distribution) is rejected. The test works best for large datasets, and if expected frequencies in bins are too small, adjacent bins should be combined to keep results reliable. The results of the Chi-Square Test used in this study can be found in Appendix A (Table a) and the square error values supporting distribution selection are documented in Appendix B (Tables b and c).

The Kolmogorov-Smirnov test

It is another method that checks if a dataset follows a specific continuous probability distribution by comparing the empirical cumulative distribution function (ECDF) to the expected theoretical CDF. Unlike the Chi-Square test, it doesn't rely on class intervals, making it more reliable for small sample sizes. The test statistic measures the largest difference between the two distributions, with a smaller value indicating a better fit. If parameters are estimated from the data, the test becomes more conservative, meaning it's less likely to reject the null hypothesis. The K-S test was used in conjunction with the Chi-Square test, and its role in determining best-fit distributions is outlined in Appendix A (Table a) and Appendix B (Tables b and c).

For thickness, a single distribution was applied across all product groups, as supported by Appendix B (Best-Fit Distributions for Thickness). The Input Analyzer results confirmed that an Exponential distribution provided the best fit, ensuring the model accurately reflects real thickness variations.

2.4 Verification & Validation

According to (Banks, 2010), verification and validation are critical steps in discrete-event simulation to ensure the model accurately represents the real-world system and provides meaningful results. It mentions that verification is about checking if the model follows the right logic, making sure that processes, resources, and interactions behave as they should. This is done by running test cases, debugging errors, and checking that everything is functioning properly. Validation is about making sure the model's outputs match real-world performance by comparing key metrics to actual data and running sensitivity analysis. Once the model is iterated and can be verified and validated, it can be used to test different scenarios. Otherwise, the results might not be meaningful and conclusions drawn from the simulation wouldn't be reliable. To ensure that the model was built correctly and produced results that aligned with expectations, both verification and validation were considered.

For verification, the focus was on checking whether the Arena model followed the conceptual model and that all processes behaved as expected. Several checks were performed to ensure this:

- The model was tested with different numbers of piles to confirm that the results remained logical.
- Input distributions were analyzed to ensure they matched the dataset from SSAB.
- Dispose modules at decision points were monitored to confirm they always stayed at zero, ensuring that no plates were lost or incorrectly removed from the system.

Validation in this case was not about comparing results to real production data, because there was none. The dataset came from another SSAB facility, but we were modeling a new one, so the main focus was on ensuring that the model met the expected KPIs and targets outlined in the project scope. Once the model was completed, there were no adjustments based on real-life feedback.

We had Teams discussions with SSAB representatives, but these were only for clarifications regarding the dataset and facility setup, not for validating the results. At the end, the simulation results matched the expectations given in the project documentation, which was enough to consider the model complete.

2.5 Experimental design and performance metrics (KPIs)

Desired goals - being Lean

The main goal is to achieve an output of 7500 tonnes of steel plates, which is set as a main KPI with the highest priority. Followed by reaching a high utilization of the oven and the pile storage, with a preferable turnover rate. What to also consider which comes almost naturally in the modern large scale industries is aspects of lean, to minimize wastes (Krafcik, 1988). Taking this in regard to aspects such as standing, tied-up capital, waiting time and utilization per section. As KPIs this would be queue size in the piles, queue size on the conveyors, crane utilization, needed facility area, down time etc (see Table 1.).

Table.1 KPI's being studied and considered used as parameters.

KPI's	Desired amount	Unit	Priority
Output: Steel plates	7500	Tonnes / week	High
Oven utilization	>90	%	High
Tied up capital *	(Low)	Volume/piles	High
Bottle necks	none	occurrences	Medium
Bottle necks	(Low)	Queue size/waiting time	Medium
Crane utilization	High and even	occurrences	Medium
Piles**	(Low)	Amount	low
Pile	< 1800	Height (mm)	Medium
Bays	< 4	Amount	low
Groups	11	Height	low
Sub-groups	1-3 / group	Length	Medium
Down time	10	hours/week	-
Distributions	Type	Height, Square error	High
Distributions	Type	Length, Square error	High
Distributions	Type	Width, Square error	High
Distributions	Type	Thickness, Square error	High

*In consideration to wanting to send as many batches of same or similar attributes to the furnace to not having changed temperature. The goal is to have a high turnover rate.

**As need for balancing sub-groups

2.6 Optimization strategy

The strategies are built around what order of the groups and sub-groups is deemed suitable to be placed. The different set-ups have their own benefits depending on the prioritization of in each case. One of the most important considerations is which crane is utilized and the relation to the limitation of the individual connected bay. Plates that were needed more often were initially placed closer to the crane to reduce travel time. At the same time, placement was adjusted to make sure no single crane was overloaded while others were underused. Different arrangements were tested to see how they affected crane movement and overall efficiency before deciding on the final setup.

2.7 Limitations of the model

The limitations of the model can be caused because of the assumptions and simplifications taken but also the limitations of the Software the limitations of the model itself are listed below, these were due to time constraints, and being first time users of Arena.

Model Limitations:

1. Piles with the same thickness group do not leave when another pile with the same thickness group is leaving. – we were unable to successfully implement this on Arena due to time limitation and the challenges created with non-dynamic length of plates in piles. Based on Assumption 1, this did not matter in the simulation.
2. Plates might not leave after a maximum of 1 week – as we simulated 1 week of production, this was not implemented in the model explicitly, but with the lower pile heights in the final model almost all should be removed naturally.
3. The number of cranes were not optimized for as we assumed 1 crane/bay for simplicity and lower number of cranes, changing the number of cranes would require manually adjusting which crane is requested and used in transport, this means that without further development, the cranes could be crossing each other's territory.
4. Furnace utilization cannot be measured directly as the furnace is modelled as a conveyor. The utilization was measured by summing 11 conveyor utilization, which was acceptable as total available hours for all were the same, and only one of them were used at a time.

Arena Limitations:

1. The biggest limitation was that we were unable to measure maximum pile length differences dynamically, which would result in the plate going to the pile that followed the max length difference in the piling rules. This was done manually, where the scenarios show how each thickness group was subsequently divided into length groups forming 24 piles – which is the number of piles in the current quenching system they have, if all length differences were to be accounted for, 43 piles would be required which is way above the current number of piles.
2. Piles had to be created manually, making it challenging to optimize for the number of piles as where each plate would go was set manually through expressions in decisions, this made it time consuming to manually increase or decrease the number of piles, and dynamic length differences could not be added, further reducing ability to optimize number of piles using Arena.
3. Inability to enter statements as code, limited what could be achieved in decisions, and required many modules to achieve simple tasks.

2.8 Documentation & Reporting

The different generations of adaptations for our model were saved throughout the building and adjustment process. The simulation software also provides after every run a data sheet of the results made by the model tracking the improvements.

3. Results

3.1 Model Conceptualization

As described in the methodology, conceptualization took place by employing the IDEF0 model structure. The main sets of activities were identified and placed in an hierarchical order, to align well and represent the quenching process and plant provided by SSAB. IDEF0 (fig.3.) provides a function model of the steps that were to be included in the operational model.

The main activity required for optimizing this case is the “Sorting” process, which in turn requires transportation to and from two separate conveyor lines. The simplified inputs, outputs, controls and resources, provide a visual to be translated into the operational model. Controls include rules and specifications, resources are mainly machinery and ultimately the goal of the system is to input plates and output quenched plates.

3.2 Collected Data

- Interrival Time Distribution
- Thickness Distribution (Chi-squared, Kolmogorov tests)
- Length, Width Distribution (made into meters for consistency of model)
- Weight Calculation → Flow Logic Chart

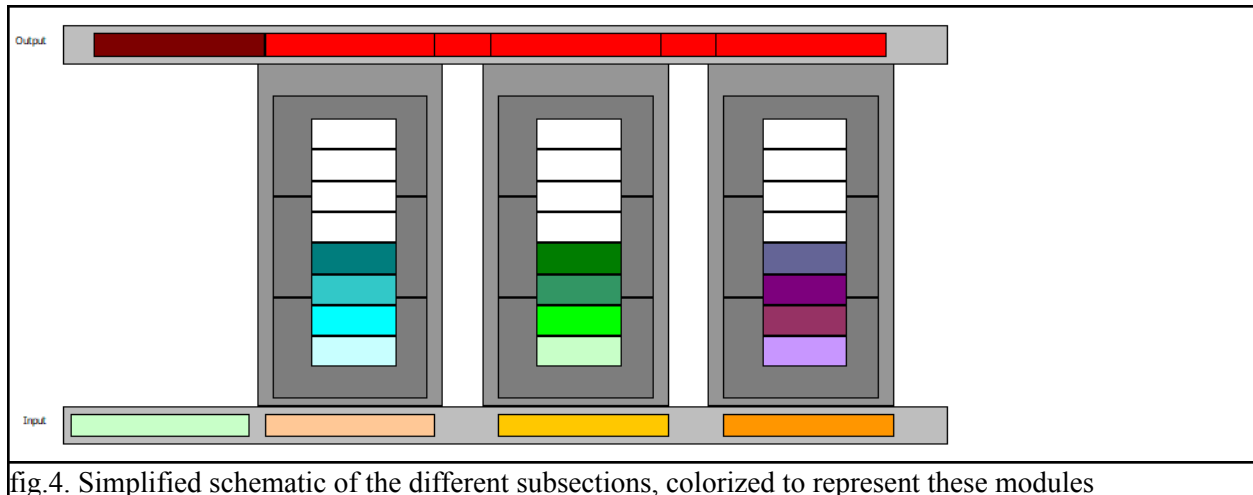
Note: Details can be found in the Appendix.

3.3 Model Translation

The model was built of modular subsections representing the different components in the system. This can be seen in the model by the enclosed sections. The intention for making these modules was to make a standardized system easy to replicate, isolate, adjust and resize. Having this access was highly useful when the simulation was to be adjusted and in the rearrangement during the preparation for a new scenario. This was also to assist the built-in modules, which were not always able to provide essential service as expected.

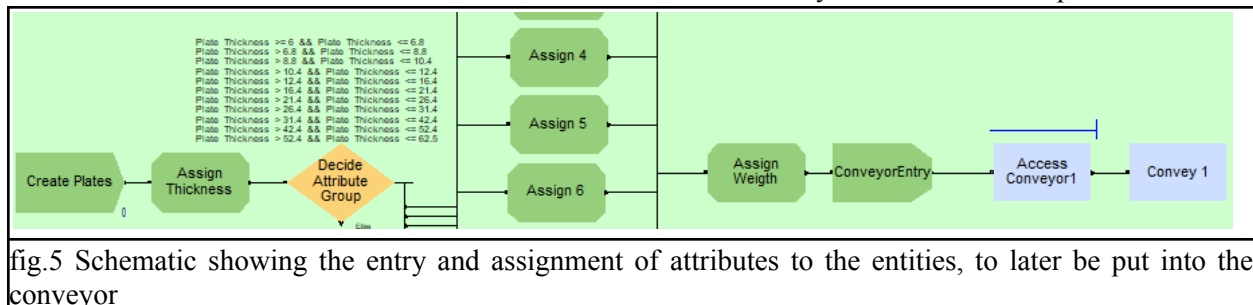
Operational Model Parts

The modular subsection consists of five different main types, these are representing the components of the plan overview that SSAB provided (see Appendix D). The types and layout are presented in the figure (fig.4.). In total 28 subsections are being utilized when considering the repeats such as those modules representing piles. The different parts are connected through material handling functions such as conveyor or by crane. This favors accessibility to repositioning of the desired visualisation besides only emulating the actual movement of goods. Each subsection consists of the software modules in an arranged way to represent our logic of the flow.



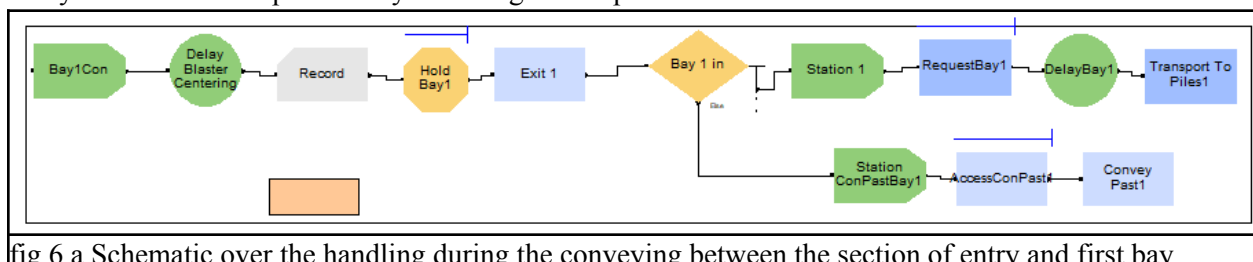
Input/attribute generator (light green)

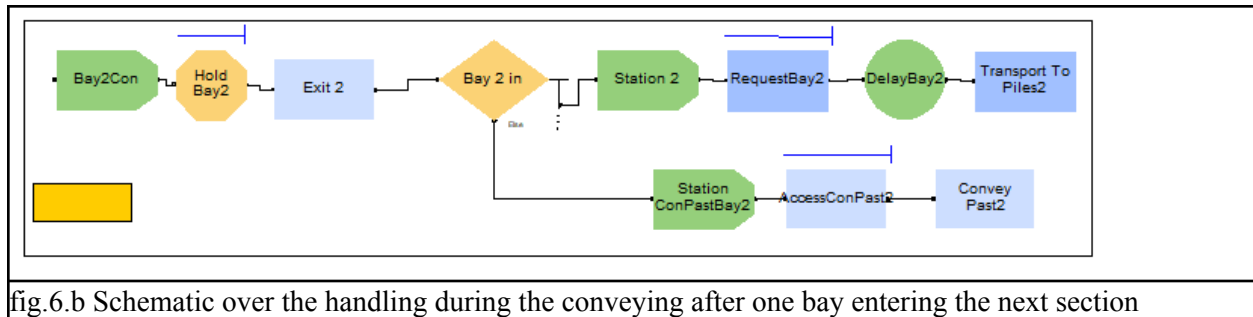
The initial part of the model (fig.5) creates the entity of plates, and follows a structure to assign required attributes to each individual plate created. This step is highly critical as it determines the input number of plates and their respective attributes. Thickness is assigned to all plates using a single distribution, where after the plates are divided into their thickness groups (1-11) after which they are assigned their length and width (e.g. Thickness Group 1 proceeds to Assign 1). Weight is then calculated using the volume and density of the steel, giving the weight in tonnes. The plates are then moved by the conveyor to the bay start. The statistical data distributions are discussed in the Data Analysis section of the report.



Conveyor to Bay, sort or pass (Orange gradient)

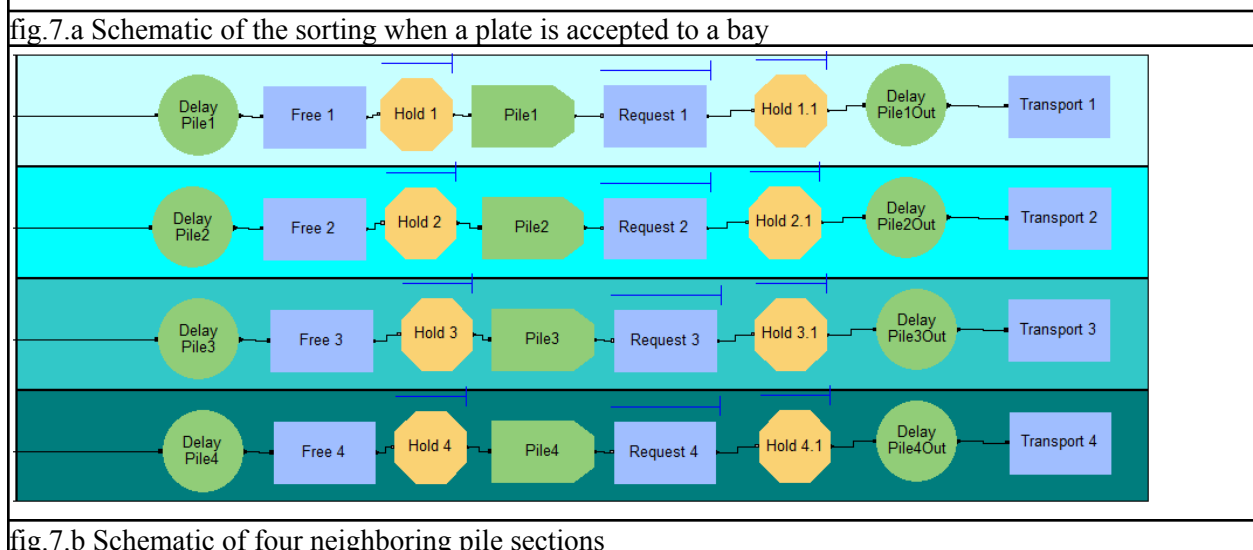
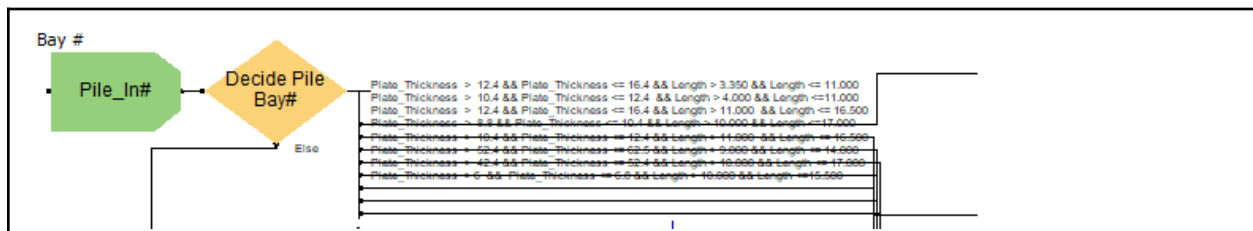
This two-part subsection controls the conveying speed between the entry and bay (fig.6.a) or to the next bay, which is done by receiving entities from earlier step by conveyor (fig.6.b), queueing up if bottleneck would have occurred and then sent to the decision function controlling at which pile section the entity should be stored and moved through. The decision is made with criterias of thickness and length of the entity and what the respective bay is willing to accept.





Pile (Blue, green or purple gradients)

When being assigned a pile (fig.7.a,c) the crane will have to be occupied with that process for the duration of movement speed times distance to and from the pile (fig.7.b,c) with the pick/drop operation time included. To represent the transport to storing the steel plates there is the delay and freeing of the crane. The piling is made by the Hold module enabling a queue to be built up. The queue is filled up until a decided threshold is reached which the Hold module is constantly checking for. When satisfied the hold is released into a request module, requesting the entities to be emptied from the pile subsection to the conveyor headed for the furnace. This is done by the same crane that would have carried in plates to the bay, which will therefore hinder it from keeping on filling up during the emptying. The reasoning behind choosing to occupy it to this extent is that it will send similar plates to the furnace at once, facilitating not needing to change temperature too often, helping to utilize the furnace which is deemed to be of higher priority. Due to limitations of the software a second holding module had to be added for the program to be able to register the request being occupied. Without this safety measure the entities started to “teleport” randomly or not signalling in the correct way according to how the release logic was set up. Teleporting could in some cases benefit travel time.



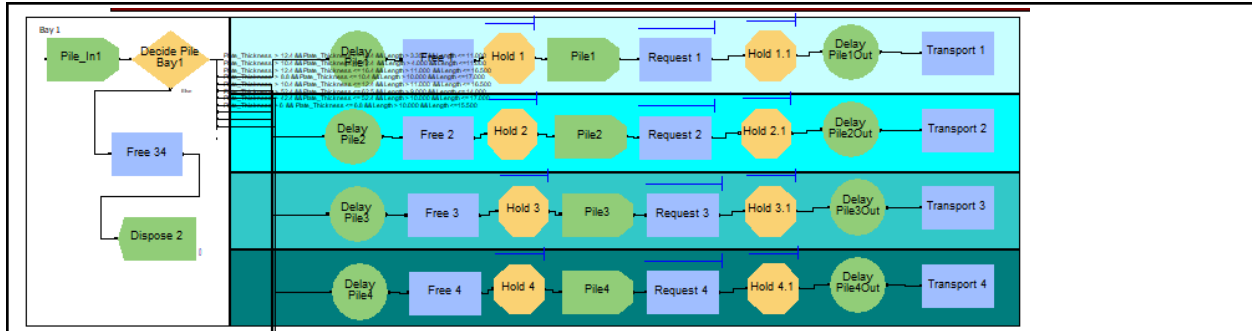


fig.7.c Schematic of sorting and piles connected

Conveyor to furnace (Red)

When a pile is being emptied it is collected in subsection representing the second conveyor (fig.8.). This is a collector for the three bays assigned with a segment setting letting the entry point have individual distances to the output. If there were to be more than one crane emptying delivering from a bay at once there will be a queue built up hindering the entities to be sent off.



fig.8 Schematic of conveyor carrying the plates into the furnace.

Furnace (Dark red)

The furnace contains several lanes of conveyors which represent the same conveyor in the real furnace(fig.9.). The choice of having more than one conveyor is due to limitations of the simulation software controlling and adjusting speed and handling several entities at the same time, meanwhile being able to record the utilization in a satisfying way. It is also not allowed to make it more compact using the sub-process module due to Arena not letting conveyors be compressed in. However, an advantage of conveyors was implementing gaps of 0.7m between plates as they enter the furnace through accessing the conveyor with Length + 0.7m on for movement.

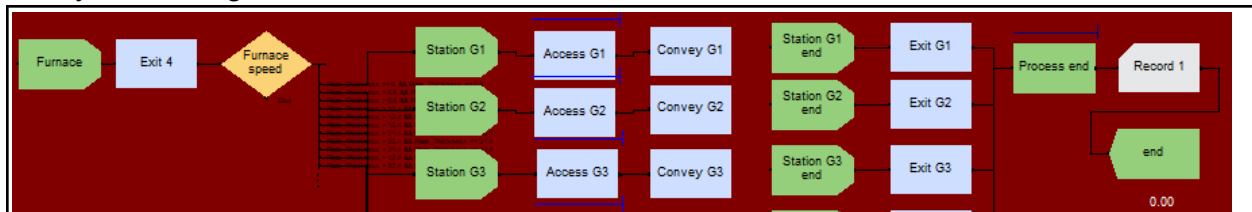


fig.9. Schematic of the process inside the furnace, showing limited for three types of settings.

Modules & Decisions

Decisions in Bay: When the plates arrive on the conveyor (rolling table) to the beginning of Bay 1, a decision module asks whether that plate belongs in that bay or not. This is done through checking the thickness group and subgroup (see Table 2). If it does belong in the bay, then the crane is requested and all incoming plates have to wait (queue) until that plate is picked up and piled. The same repeats for each bay.

Decisions in Piling: Once the crane moves the plate from the conveyor (rolling table), another decision takes place with similar conditions, checking for subgroups. Then the pile is assigned and the crane moves to that pile location (station), the distances between the crane initial station and the destination station are entered in the model as is the crane speed (averaged 1m/s), this determines the sideways movement time taken.

Table. 2 The expressions used inside the decide modules based on allocation of groups in bays & piles

Expression	Group
Plate Thickness > 6 && Plate Thickness <= 6.8 && Length > 3.400 && Length <=10.000	1
Plate Thickness > 6 && Plate Thickness <= 6.8 && Length > 10.000 && Length <=15.500	
Plate Thickness > 6.8 && Plate Thickness <= 8.8 && Length > 3.400 && Length <=10.000	2
Plate Thickness > 6.8 && Plate Thickness <= 8.8 && Length > 10.000 && Length <=16.000	
Plate Thickness > 8.8 && Plate Thickness <= 10.4 && Length > 3.100 && Length <=10.000	3
Plate Thickness > 8.8 && Plate Thickness <= 10.4 && Length > 10.000 && Length <=17.000	
Plate Thickness > 10.4 && Plate Thickness <= 12.4 && Length > 4.000 && Length <=11.000	4
Plate Thickness > 10.4 && Plate Thickness <= 12.4 && Length > 11.000 && Length <= 16.500	
Plate Thickness > 12.4 && Plate Thickness <= 16.4 && Length > 3.350 && Length <= 11.000	5
Plate Thickness > 12.4 && Plate Thickness <= 16.4 && Length > 11.000 && Length <= 16.500	
Plate Thickness > 16.4 && Plate Thickness <= 21.4 && Length > 4.000 && Length <= 10.000	6
Plate Thickness > 16.4 && Plate Thickness <= 21.4 && Length > 10.000 && Length <= 16.500	
Plate Thickness > 21.4 && Plate Thickness <= 26.4 && Length > 4.000 && Length <= 10.000	7
Plate Thickness > 21.4 && Plate Thickness <= 26.4 && Length > 10.000 && Length <= 17.000	
Plate Thickness > 26.4 && Plate Thickness <= 31.4 && Length > 4.000 && Length <= 9.000	8
Plate Thickness > 26.4 && Plate Thickness <= 31.4 && Length > 9.000 && Length <= 15.000	
Plate Thickness > 31.4 && Plate Thickness <= 42.4 && Length > 3.700 && Length <= 9.000	9
Plate Thickness > 31.4 && Plate Thickness <= 42.4 && Length > 9.000 && Length <= 16.500	
Plate Thickness > 42.4 && Plate Thickness <= 52.4 && Length > 3.000 && Length <= 10.000	10
Plate Thickness > 42.4 && Plate Thickness <= 52.4 && Length > 10.000 && Length <= 17.000	
Plate Thickness > 52.4 && Plate Thickness <= 62.5 && Length > 3.000 && Length <= 9.000	11
Plate Thickness > 52.4 && Plate Thickness <= 62.5 && Length > 9.000 && Length <= 14.000	

Warm-up and Replications

A warm up period of 5 days was chosen for the simulation based on the insight from SSAB, to ensure that there were 3000 tons of plates piled up at the start of the new week, creating a steady state. Based on this the simulation time is for 12 days, giving 7 days of statistics from the simulation. 5 days was decided through trial and error and observing the total weight in and out of the system, in this case the output was irrelevant as long as ,weight in minus weight out, reached 3000 tons, which happened around the 5 day mark. A different approach considered was to hold the pile infinitely (using the hold module) until the weight in was recorded as 3000 tons, however this meant that pile heights were above the 1800mm at the start of the new week. This would create an unrealistic and unsafe scenario therefore the first approach described was followed.

The simulation time per day was set to 23 hours to account for breaks taken by workers etc. This gave 7 hours of breaks every week, however this does not create total randomness as the model uses each hour all at once rather than (e.g. 4x 15 mins breaks/day) or a longer break/maintenance break throughout the week. This can be considered a weakness of the model, but to the best of our ability we were unable to create these breaks at random when using Arena.

4. Findings

4.1 Preparing

To enable building the model key components had to be identified out of the given data regarding the case, this included the historical data about the steel plate products and the attributes of the different operating machines such as crane speed or allowed positioning of plates.

Data analysis

Data collection was performed by specialists at SSAB, therefore our initial step was to clean the data, then identify the distribution, choose parameters, and finally, evaluate for goodness of fit. The data cleaning was performed based on the restrictions given by SSAB. Since the data was historical data from an old system, thickness less than 6mm was cleaned based on the information we gained, similarly any outliers and missing data points were also cleaned. More than 17,000 data points were removed by analysing the thickness column, however given the vast amount of data, the historical data was still applicable. Moving on the cleaned data was used in further analysis, and Arena Input Analyzer was used throughout (Fig.10.). Following the structure of the conceptual model, the interarrival time distribution was determined first. The distribution was found as $-0.001 + \text{EXPO}(2.9)$ minutes. Given the insignificance of -0.001 , $\text{EXPO}(2.9)$ was used in the model, a plate arriving on average every 2.9 minutes following a randomly distributed exponential form. A square error of 0.000368 shows that the difference between the fitted distribution and actual data is low.

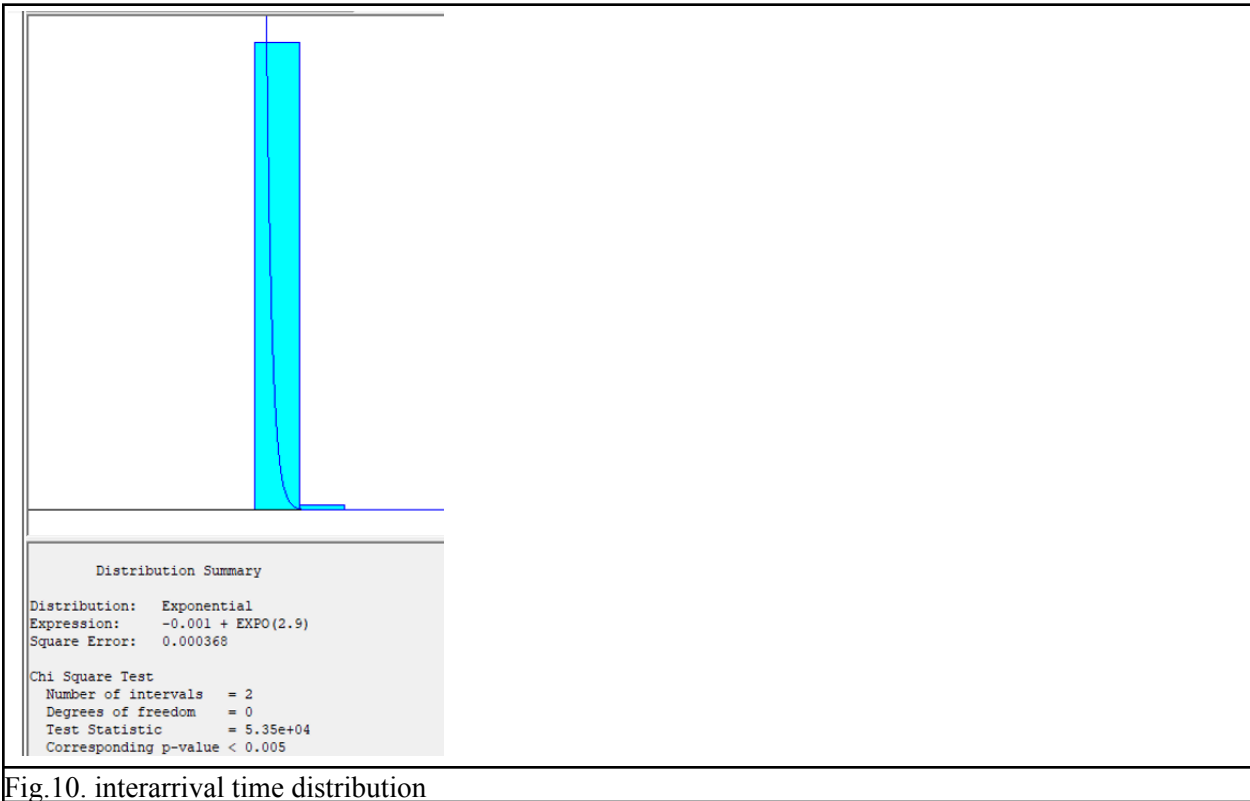
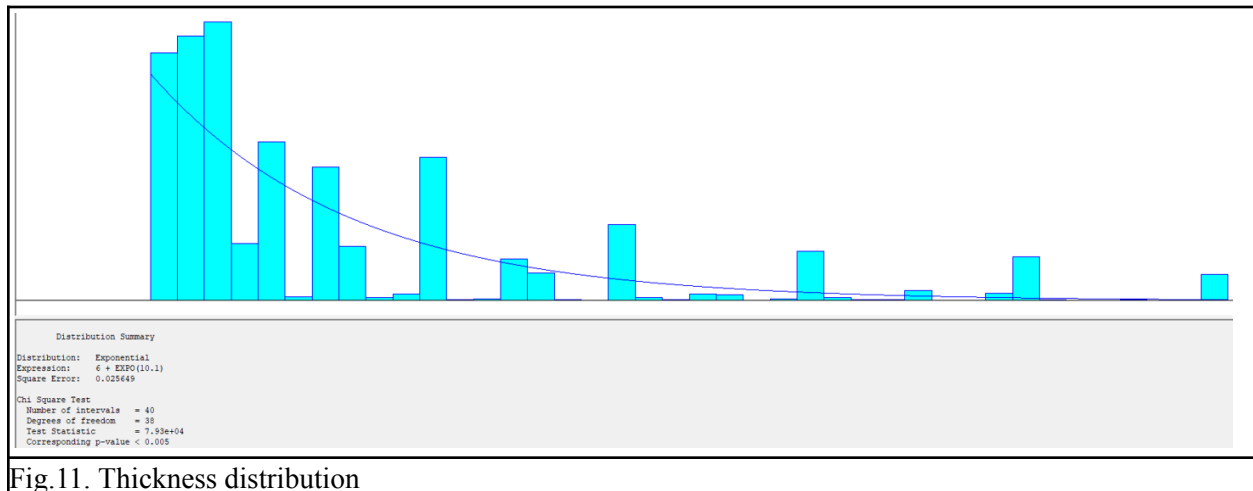


Fig.10. interarrival time distribution

Next the thickness distribution was determined using all group values, creating a single probability distribution for all plates (Fig.11.). This approach helped maintain randomness and real world variability, keeping the uncertainty in the plate thickness of the created entity. Having separate distributions would also require separate interarrival time distributions, this would assume fixed arrival proportions for each

thickness group, reducing randomness. With our approach of single interarrival time distribution along with a unified thickness distribution we hoped to avoid overfitting historical data.



The number 6 in the distribution refers to the minimum thickness + the exponentially distributed assignment of the thicknesses. The chi-squared results of 7.93×10^4 , and p-value < 0.005, might indicate that the distribution is not the best fit. However, for practical reasons, the exponential model effectively captures the natural trend where thinner plates are more common than thicker ones. Since the simulation performance is more dependent on overall behavior than exact statistical fit, this approach remained valid for our model. A maximum plate thickness was incorporated into the distribution to ensure no extremely thick outlier plates were created. Individual group's thicknesses were also analysed, all providing a p-value < 0.005, therefore there was no perfect fit, and we chose to capture the randomness, without further dividing the groups into subgroups to get a better fit.

The length and the width distributions were assigned based on thickness groups. This meant that 22 different distributions were created using Arena Input Analyzer. The histograms of each distribution were observed to see whether the given distribution is actually appropriately fitting. The exact distributions can be seen in Appendix B. These distributions were converted into meters from millimeters to keep consistency over the model and ensure they take up the correct spacing on the conveyors etc. The fits were checked again to ensure they fit in the same way with the data in meters, a couple of distributions were changed slightly for use in the model. Similarly maximum lengths were set to some distributions based on the historical data to avoid extremely long and unrealistic attributes created (e.g. 30m long plates), which happened to be only a couple of plates that were being disposed, this solved it.

Input analyzer showed some problems where similar to the thickness distributions, several peaks existed in most data groups. For example in Group 4 width (see fig.11.) there were 3-4 peaks, but only one is well accounted for in the distribution given. Problematic fits like these were used in the model, but given the simplifications that were made due to Arena's limitations, the width and length did not create much difference in the piling as SSAB intended since groups were divided into piles as having 2 groups of lengths and not dynamically. The major contribution it had was on the weight calculation, which determines the overall output tonnage, therefore can be determined as the model's weakness. Dividing these data into sub groups was considered, but rejected given the time limitations, the lack of ability to add dynamic pile assignment based on lengths (making the actual length information less valuable to us in optimizing) and the need for many "decisions" in Arena to implement this.

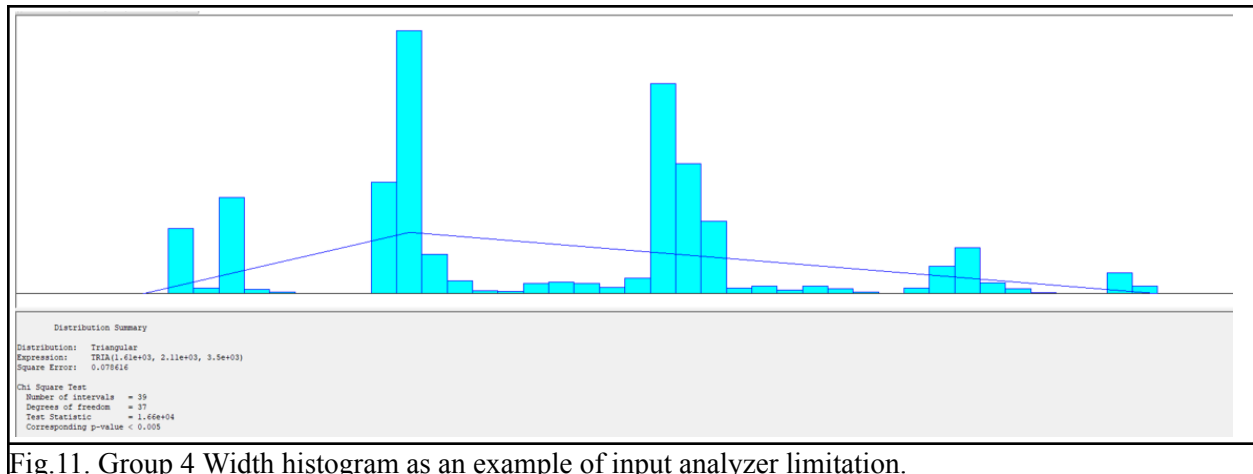


Fig.11. Group 4 Width histogram as an example of input analyzer limitation.

Setting up the model

By applying the logic of our already structured flowcharts together with the findings in the data analysis into the software it was possible to initiate the first test run. This was made in a less extended imitation of the total planned model, just implementing a handful of piles to be able to identify immediate potential obstacles for the flow. By using our modular system we could thereafter expand this to the desired amount of starting piles to reach a model to be ready for scenario testing.

Adjustments

Several iterations of adjustments were to be implemented. In the initiation of the set-up it was found that the simulation software lacked in basic module functions expected to supply. Throughout the whole modelling there were occasional minor obstacles that had to be resolved, with no major issues. This could include routing distances for the cranes to the piles. Also to compensate for the furnace settings there had to be adaptations with added lanes.

4.2 Scenarios and change of parameters

When the simulation set-up was well adjusted to handle the data representing the input and being able to send through the entities in a satisfying flow the scenarios were constructed. The main focus was put into pile positioning and dimensions of the piles. Because of the different frequent occurrences of the categories of plates, aspects such as distance traveled so that the most common should be handled first, or distributing of the frequencies in the piles so that the cranes were equally utilized. When the pile positioning was deemed to be satisfied it was identified that during this simulated working week several piles were just a few plates from transitioning to the emptying phase. The release is executed when the height threshold is passed. Due to the amount of deliveries being discrete and rounded down, almost filled piles locked up a significant volume of steel plates. The original threshold was at the maximum allowed height limit of a pile (1800mm), however by lowering this threshold the stuck groups were able to be emptied, which improved the flow of tied up capital in material articles. By releasing locked in plates, the final output was able to be adjusted to be 8 tons off from the desired goal. Approach Summary: First, assign each thickness group and its respective length group to the best bay location. Second, determine the best pile height for each pile. The following seven scenarios describes the findings of changing the parameters (data tables can be found in Appendix E-F):

Scenario 1, Baseline: Each bay was assigned groups divided from the given groups of 1 to 11 distributed into piles, with bay 1 having 6 piles, and bay 2 & 3 having 8 piles each. This gave each group two piles, divided as short and long plates (according to the length ranges). The pile height threshold was kept at 1750mm, allowing a buffer of 50mm to the maximum 1800mm to make sure it never exceeds the maximum (See Appendix E). The baseline scenario gave a promising result, but off from the desired.

Output: **7101** (tons/week)
Furnace utilization: **85.00** %

Scenario 2, Piles ordered by group and subgroup in ascending order: By positioning the piles in ascending order of total pile heights within the results of a week simulation (e.g. pile 5 had 12064.17 total height equalling 6.89 disposals) from Scenario 1. This gave worse results than scenario 1, bottleneck (queue) at the start of bay entry. The reasoning was that because of some entity-groups being more popular they should be prioritized to be picked as soon as possible, freeing up the conveyor. However, that led to a major bottleneck due to the first crane being occupied emptying piles while the others waited to be stored, and the other bays were barely utilized (See Appendix F)

Scenario 3, Piles ordered by group and subgroup in descending order: Similar to scenario 2, however bay 1 and bay 3 were switched. It gave better results compared to scenario 2. More of the conveyor being utilized but less bottleneck in bay 1 to bay 3 movement. The pile heights were kept the same. Still worse than the reference baseline scenario (See Appendix E).

Output: **7034** (tons/week)
Furnace utilization: **87.13** %

Scenario 4: Even Distribution of Thickness Across Bays: Changed bay assignment of piles and not the pile height. The reasoning is, if there was an even amount of total pile heights in each bay, each crane should be utilized the same. The results were improved compared to scenario 3, with a more equally distributed workload over the bays. Still less output than the reference baseline scenario, but with the highest furnace utilization as of yet measured (See Appendix E).

Output: **7096** (tons/week)
Furnace utilization: **89.50** %

Scenario 5: Even Thickness Distribution with Rearranged Order: Based on Scenario 4, modified from observing bottleneck piles in each bay and moving them to different bays, while keeping the balance, taking into account the remaining plate weights in each pile at the end of the week. Gave slightly better results than scenario 4 and more equally distributed workload. (See Appendix E).

Output: **7110** (tons/week)
Furnace utilization: **85.58** %

Scenario 6: Even Thickness Distribution with Adjusted Pile Height: Changes pile height not bay assignment of piles. Based on Scenario 5, leftover pile heights at the end of the week were observed, and the pile heights changed to **1 meter** to test for improvements. Gave greatly better results than scenario 5, less leftover plates in piles and higher furnace utilization (See Appendix E).

Output: **7404** (tons/week)
Furnace utilization: **86.81** %

Scenario 7: Final, Even Thickness Distribution with further Adjusted Pile Height: Based on Scenario 6, leftover pile heights at the end of the week were observed, and the pile heights changed. Gave better results than scenario 6, and the best overall. There were less leftover plates in the piles, and it ensured plates would leave after 1 week, since they are assigned lower pile heights. (See Appendix E). The output became about 0.1% above the desired amount with around 95% utilization of the furnace and deemed to be satisfying.

Output: 7508 (tons/week)

Furnace utilization: 94.86 %

Scenario Comparison of Output and Furnace Utilization

To better illustrate the impact of the different scenario optimizations, the figure below(fig.12) provides a visual summary of the output (blue bars) and furnace utilization (orange line) across all tested cases. The initial scenarios (1-5) show relatively small variations, with Scenario 2 slightly underperforming due to bottlenecks. Scenario 6 marks a significant improvement, and Scenario 7 achieves the highest performance, with an output of 7508 tons per week and a furnace utilization of 94.86%.

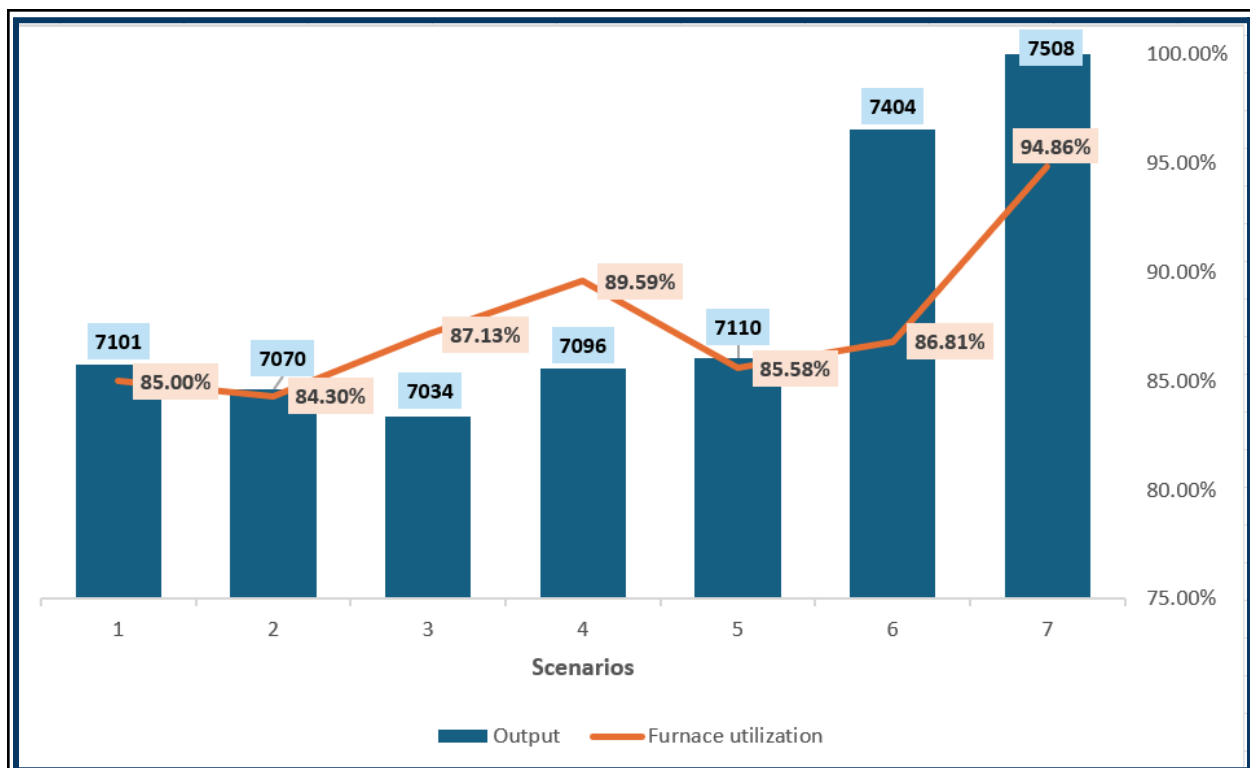


Figure 12: Scenario Performance: Output and Furnace Utilization Trends

5. Discussion

Several discussions can be drawn from the optimization approach and process. In relation to RQ1, it was observed that the pile height affects the tonnes processed per week significantly more than the pile location. For example when changing the pile location the tonnage per week was always around 7000-7100 tons, showing a small improvement reaching from Scenario 1 - Baseline to Scenario 5, even though scenario 2 to 4 showed decrease relative to the Baseline. Regarding the furnace utilization, related to RQ2 and RQ3, it saw its biggest improvements when the piles were adjusted so that the bays had an equal distributed flow. Therefore the bottlenecks were one of the major contributors to utilization level, and can be seen further throughout the discussion.

Scenario 1 to scenario 2, showed that having the frequently moving piles in bay 1, for shorter conveyor travel distance, did not present improvements but a slight worsening in output. This was a result of bottlenecks being created at the start of the first bay, as the crane frequently had to empty piles towards the furnace. Scenario 2 to scenario 3, was the same case but reversed, while it improved upon scenario 2, it was still worse than scenario 1. This showed that the baseline was a good starting point, with the ordered groups in piles showing a good average crane utilization.

Therefore going from scenario 3 to scenario 4, the bays were aimed to have similar total plate thicknesses throughout the week, aiming to equally distribute time and workload between the three cranes. This was still worse than the baseline, but showed improvements in comparison to scenario 3, proving that even work load between cranes was favourable. Last change of pile bay assignment occurred from scenario 4 to scenario 5, where observations such as bottlenecks and crane movements led to switching of some piles to create better workload distribution, which improved the output by a little.

At scenario 6, we tested a pile height of 1000mm for all piles, this led to a jump from around 7100 tons to 7400 tons, showing the significance of pile height, as smaller heights ensure that non-popular piles also leave for the furnace. This proved that the pile height made the most significant difference in reaching the KPI, therefore for the final scenario (scenario 7), the less frequently full piles were all observed at the end of each week, and the pile height reduced to an arbitrary number 300mm. This concluded our scenarios as the goal of 7500 tons/week was reached, and time limitations did not allow further scenario creation.

More improvements could be made to the pile heights, as in theory each pile could have its own perfect height to optimize for. This would likely lead to higher production which we would like to avoid as overproduction is not necessarily good if not needed. However, the furnace utilization can be improved by changing the pile heights and by controlling the stable input flow to it. Depending on the balance between the importance of the output versus utilization, for example due to energy costs, it could be good to consider this slight overproduction. In any case, scenario 7 proved to be fairly close to the desired goal.

When performing longer runs (e.g. 1-2 months of production), there was a decrease in the output to around 6500 tons/week, this also demonstrates a limitation of the model as bottlenecks are created in really long runs, but due to the limitations of the model and Arena, knowing the source of this is highly challenging at this stage. Another limitation of the model was that replication lengths were set to 1 for each scenario for computational time restraints and scenario trials, meaning that if more replications were to be made, some replications would yield higher results and some lower. However Arena runs

simulations with a confidence interval of 95% so we can expect the true mean of the simulation metrics to fall 95% of the time if the experiment were repeated in replications.

There are things we would have done differently, if this project were to be repeated. Firstly, analyzing the data from the start, and fully understanding the implications of each distribution, histogram and the goodness-of-fit results, would significantly reduce back and forth time taken in implementing these while completing the model. Garbage in - garbage out approach certainly has a significant truth to it, and noticing that every small decision/simplification has an impact is important. To aid with this we would have performed more sensitivity analysis in determining how small changes in input can affect the output, to gain an understanding of the implications of our changes & decisions. Secondly, testing parts of the model before implementing a larger model would also aid with using the time effectively and take us further, helping with verification. Furthermore, with more time, we would have made use of the expert knowledge available to validate our model given that there was no output data shared for the historical data we were given, making it challenging to validate our model.

As per the model, with more time, we would attempt to create the maximum length difference calculations per pile before a plate is assigned to a pile. This would require recording the current minimum and maximum plate length in a pile and then calculating if the incoming plate fits within the maximum length difference for that thickness group. This was attempted already but proved to be very tricky to achieve as lots of attributes needed to be checked and then calculated and implemented in the decision module, ultimately we could not implement it. We would also have made use of sub-models to represent different parts of the model, to make it easier to work with and visually more appealing.

Finally, if given the choice we would choose a software that accepts code as input for decision making, which would significantly reduce the time taken, and also help create a more dynamic and realistic model.

Conclusions

To conclude, with the discussions made, the assumptions and simplifications taken, and the model limitations, the output of 7500 tons was reached (RQ1). The simulation created in the Arena software was able to demonstrate a scenario that performed satisfactorily in relation to the desired goal within the scope set for this project. With a total output of 7508 tons of steel plates per week, or about 0.1% more than the target, and with furnace utilization at 95%, the furnace can be considered to be running almost constantly (RQ2). The improvements were seen during the adjustments to balancing pile height and distribution to the different bays according to the frequency of occurrence of the different entities (RQ3).

The simulation was built based on the data we had access to and how we interpreted it, but it could not be validated through real-world application. This does not guarantee that the real-world performance would be the same, and considering the limitations of both Arena and the model, the results should not be taken at face value without further consideration of practical implications

References

Banks, J. (2010) *Discrete-event System Simulation*. Pearson.

Fechter, M. and Seeber, C. (2018) ‘Integrated Process Planning and Resource Allocation for Collaborative Robot Workplace Design’, *ResearchGate* [Preprint]. Available at: <https://doi.org/10.1016/j.procir.2018.03.179>.

Historia (2024) *SSAB*. Available at: <https://www.ssab.com/sv-se/ssab-koncern/om-ssab/ssab-i-korthet/history> (Accessed: 2 December 2024).

Krafcik, John.F. (1988) ‘Triumph of the Lean Production System’.

Sild, S. (2023) ‘Quenching Explained – Definition, Process, Benefits and More’, *Fractory*, 24 March. Available at: <http://https%253A%252F%252Ffractory.com%252Fquenching-explained%252F> (Accessed: 8 March 2025).

SSAB (2023) *SSAB Årsredovisning 2022*. Annual report. Stockholm. Available at: [ssab.se](https://www.ssab.se).

Appendix

Appendix A

Table (a). The table summarizes the best-fit probability distributions for each key parameter group, generated using Arena Input Analyzer. Best-fit distribution determined using Chi-Square Test and Kolmogorov-Smirnov Test, ensuring alignment with real production patterns.

Group	thickness	Length	width	time	weight
1	6 + LOGN(0.683, 34.7)	TRIA(3.42e+03, 1.23e+04, 1.55e+04)	TRIA(1.68e+03, 2.6e+03, 3.32e+03)	-0.001 + EXPO(2.9)	328 + 2e+03 * BETA(4.39, 5.85)
2	TRIA(6.71, 7.99, 8.99)	TRIA(3.4e+03, 1.22e+04, 1.58e+04)	TRIA(1.69e+03, 2.6e+03, 3.46e+03)		412 + 2.42e+03 * BETA(4.13, 4.76)
3	8.75 + 1.8 * BETA(7.75, 4.37)	TRIA(3.11e+03, 6.16e+03, 1.59e+04)	1.33e+03 + LOGN(1.12e+03, 491)		TRIA(371, 1.14e+03, 3.6e+03)
4	10.3 + WEIB(1.73, 5.06)	4e+03 + LOGN(4.77e+03, 4.83e+03)	TRIA(1.61e+03, 2.11e+03, 3.5e+03)		654 + GAMM(706, 1.6)
5	NORM(15.3, 0.727)	3.35e+03 + LOGN(5.08e+03, 3.38e+03)	1.36e+03 + LOGN(1.05e+03, 470)		448 + LOGN(1.86e+03, 1.07e+03)
6	16 + WEIB(4.17, 13.8)	TRIA(4e+03, 5.96e+03, 1.45e+04)	1.59e+03 + LOGN(829, 430)		942 + LOGN(2e+03, 1.41e+03)
7	22 + WEIB(3.25, 9.01)	TRIA(4e+03, 5.97e+03, 1.45e+04)	TRIA(1.68e+03, 2.11e+03, 3.5e+03)		1.44e+03 + LOGN(2.28e+03, 1.94e+03)
8	NORM(30.3, 0.409)	TRIA(4e+03, 6.06e+03, 1.44e+04)	1.68e+03 + ERLA(144, 4)		1.7e+03 + LOGN(2.54e+03, 1.98e+03)
9	TRIA(31, 39.8, 43)	3.52e+03 + 1.08e+04 * BETA(1.94, 1.36)	TRIA(1.68e+03, 2.09e+03, 3.41e+03)		NORM(6.95e+03, 1.86e+03)
10	42 + WEIB(8.22, 5.63)	TRIA(3e+03, 6.42e+03, 1.39e+04)	1.67e+03 + GAMM(177, 3.12)		2.35e+03 + WEIB(5.69e+03, 2.92)
11	52 + WEIB(8.21, 18.9)	TRIA(3e+03, 6.42e+03, 1.39e+04)	1.65e+03 + 1.54e+03 * BETA(2.96, 4.19)		TRIA(3.77e+03, 1.12e+04, 1.28e+04)
12	6 + EXPO(10.1)	3e+03 + 1.29e+04 * BETA(0, 0)	1.33e+03 + 2.18e+03 * BETA(4.61, 4.49)		328 + LOGN(2.19e+03, 1.91e+03)

Appendix B

This appendix summarizes the process of fitting probability distributions to the collected data. The best-fit distributions were chosen based on statistical tests such as the Chi-Square Test and Kolmogorov-Smirnov Test. Square error values were also used to assess how well each distribution matches the actual data.

Table (b). Best-Fit Distributions for **Length**: best-fit probability distributions for plate length across different groups. Distributions chosen based on statistical tests to make sure they match real production data as closely as possible. The square error values indicate how well each distribution fits the data

Group	Best-Fit Distribution (Length)	Parameters (Length)	Square Error (Length)
1	Triangular	3.42e+03, 1.23e+04, 1.55e+04	0.060046
2	Triangular	3.4e+03, 1.22e+04, 1.58e+04	0.066295
3	Triangular	3.11e+03, 6.16e+03, 1.59e+04	0.16028
4	Lognormal	4e+03 + LOGN(4.77e+03, 4.83e+03)	0.171654
5	Lognormal	3.35e+03 + LOGN(5.08e+03, 3.38e+03)	0.170131
6	Triangular	4e+03, 5.96e+03, 1.45e+04	0.178711
7	Triangular	4e+03, 5.97e+03, 1.45e+04	0.117683
8	Triangular	4e+03, 6.06e+03, 1.44e+04	0.158243
9	Beta	1.94, 1.36, Scale: 1.08e+04, Shift: 3.52e+03	0.036101
10	Triangular	3e+03, 6.42e+03, 1.39e+04	0.05251
11	Triangular	3e+03, 6.42e+03, 1.39e+04`	0.05251

Table (c). The best-fit probability distributions for plate **width** across all groups. Similar to length, these distributions were selected based on statistical testing, ensuring that the model accurately captures the variability in plate width. Square error values were used to assess the fit of each distribution.

Group	Best-Fit Distribution (Width)	Parameters (Width)	Square Error (Width)
1	Triangular	1.68e+03, 2.6e+03, 3.32e+03	0.043441
2	Triangular	1.69e+03, 2.6e+03, 3.46e+03	0.054622
3	Lognormal	1.12e+03, 491, Shift: 1.33e+03	0.121932
4	Triangular	1.61e+03, 2.11e+03, 3.5e+03	0.078616
5	Lognormal	1.05e+03, 470, Shift: 1.36e+03	0.098475
6	Lognormal	829, 430, Shift: 1.59e+03	0.12552
7	Triangular	1.68e+03, 2.11e+03, 3.5e+03	0.107313
8	Erlang	144, 4, Shift: 1.68e+03	0.117099
9	Triangular	1.68e+03, 2.09e+03, 3.41e+03	0.046264
10	Gamma	177, 3.12, Shift: 1.67e+03	0.062664
11	Beta	2.96, 4.19, Scale: 1.54e+03, Shift: 1.65e+03	0.03518

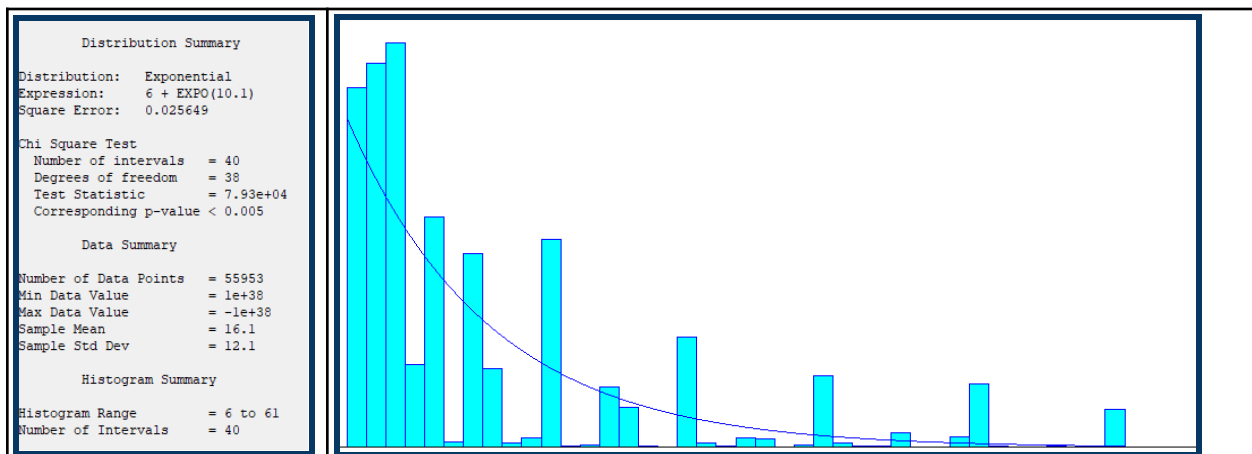


Fig. B. Input Analyzer Result: The thickness distribution applies to all product groups. The best fit for thickness data is an Exponential distribution ($6 + \text{EXP}(10.1)$), determined through statistical tests. The square error for this fit is 0.025649, ensuring the distribution accurately reflects the variation in thickness measurements.

Appendix C

Table (d). Interarrival Time of Plates: Plates arrive at the facility following a random exponential process. This helps simulate the natural variation in production flow.

Parameter	Value
Arrival Type	Random (Exponential)
Distribution	EXPO(2.9)
Mean Interarrival Time	2.9 minutes
Units	Minutes
Entities per Arrival	1
Max Arrivals	Infinite

Appendix D

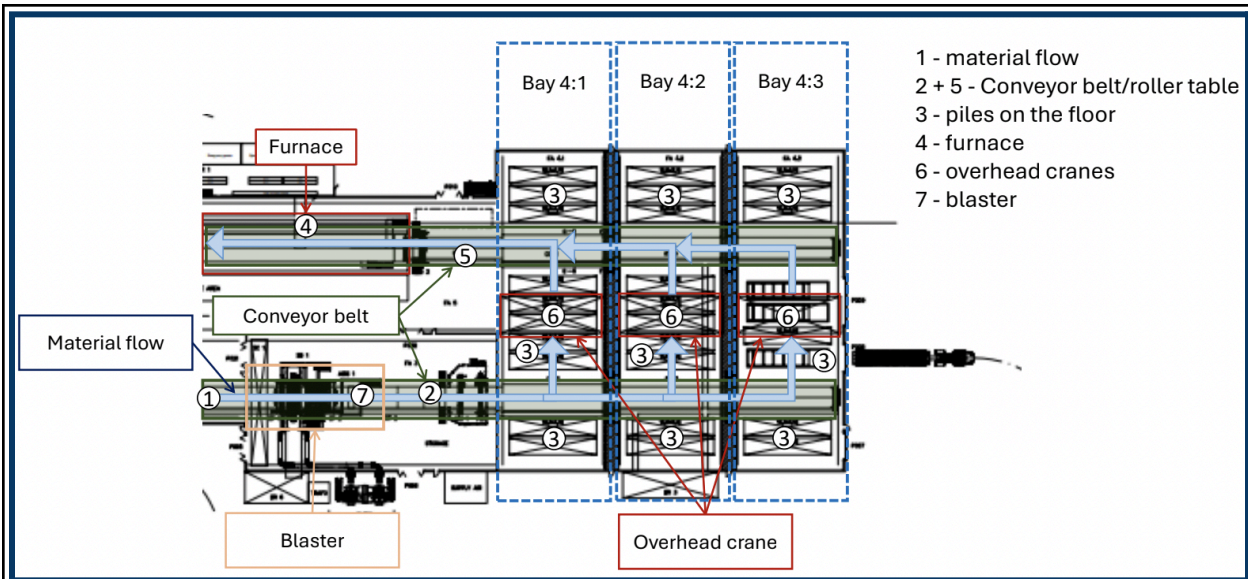


Figure D: SSAB Oxelösund Facility Layout, illustrating the key components involved in the steel plate processing flow. It highlights the movement of materials and resources, which served as the foundation for developing the Arena simulation model. Plates move through the facility following the material flow (1), guided by conveyor belts and roller tables (2, 5). Temporarily stored in piles (3) before reaching the furnace (4) for quenching. Overhead cranes (6) transport plates between bays, while the blaster (7) is used for surface treatment.

Appendix E: Scenarios 1, 3-7.

Scenario 1

Group (Length Range)	Pile Height	Pile	
Group 1 (3400-10000)	1750	Pile 1	Bay 1
Group 1 (10000-15500)	1750	Pile 2	
Group2 (3400-10000)	1750	Pile 3	
Group2 (1000-16000)	1750	Pile 4	
Group 3 (3100-10000)	1750	Pile 13	
Group 3 (10000-17000)	1750	Pile 14	
Group 4 (4000-11000)	1750	Pile 5	Bay 2
Group 4 (11000-16500)	1750	Pile 6	
Group 5 (3350-11000)	1750	Pile 7	
Group 5 (11000-16500)	1750	Pile 8	
Group 6 (4000-10000)	1750	Pile 17	
Group 6 (10000-16500)	1750	Pile 18	
Group 7 (4000-10000)	1750	Pile 19	Bay 3
Group 7 (10000-17000)	1750	Pile 20	
Group 8 (4000-9000)	1750	Pile 9	
Group 8 (9000-15000)	1750	Pile 10	
Group 9 (3700-9000)	1750	Pile 11	
Group 9 (9000-16500)	1750	Pile 12	
Group 10 (3000-10000)	1750	Pile 21	
Group 10 (10000-17000)	1750	Pile 22	
Group 11 (3000-9000)	1750	Pile 23	Bay 3
Group 11 (9000-14000)	1750	Pile 24	

Scenario 3:

Group (Length Group)	Pile Height	Pile	
Group 11 (3000-9000)	1750	Pile 1	Bay 1
Group 3 (10000-17000)	1750	Pile 2	
Group 1 (10000-15500)	1750	Pile 3	
Group 4 (11000-16500)	1750	Pile 4	
Group 11 (9000-14000)	1750	Pile 13	
Group 1 (3400-10000)	1750	Pile 14	
Group 10 (10000-17000)	1750	Pile 15	Bay 2
Group 9 (3700-9000)	1750	Pile 5	
Group 6 (10000-16500)	1750	Pile 6	
Group 10 (3000-10000)	1750	Pile 7	
Group2 (3400-10000)	1750	Pile 8	
Group 8 (9000-15000)	1750	Pile 17	
Group 5 (11000-16500)	1750	Pile 18	Bay 3
Group 7 (10000-17000)	1750	Pile 19	
Group 5 (3350-11000)	1750	Pile 9	
Group 6 (4000-10000)	1750	Pile 10	
Group 7 (4000-10000)	1750	Pile 11	
Group 9 (9000-16500)	1750	Pile 12	
Group 8 (4000-9000)	1750	Pile 21	
Group 4 (4000-11000)	1750	Pile 22	
Group2 (10000-16000)	1750	Pile 23	Bay 3
Group 3 (3100-10000)	1750	Pile 24	

Scenario 4:

Group (Length Group)	Pile Height	Pile	
Group 5 (3350-11000)	1750	Pile 1	Bay 1
Group 4 (4000-11000)	1750	Pile 2	
Group 9 (3700-9000)	1750	Pile 3	
Group 7 (10000-17000)	1750	Pile 4	
Group 8 (9000-15000)	1750	Pile 13	
Group 3 (10000-17000)	1750	Pile 14	
Group 1 (10000-15500)	1750	Pile 15	Bay 2
Group 1 (3400-10000)	1750	Pile 16	
Group 6 (4000-10000)	1750	Pile 5	
Group 8 (4000-9000)	1750	Pile 6	
Group2 (1000-16000)	1750	Pile 7	
Group 5 (11000-16500)	1750	Pile 8	
Group 11 (3000-9000)	1750	Pile 17	Bay 3
Group 4 (11000-16500)	1750	Pile 18	
Group 11 (9000-14000)	1750	Pile 19	
Group 10 (10000-17000)	1750	Pile 20	
Group 7 (4000-10000)	1750	Pile 9	
Group 9 (9000-16500)	1750	Pile 10	
Group 3 (3100-10000)	1750	Pile 11	
Group 10 (3000-10000)	1750	Pile 12	
Group2 (3400-10000)	1750	Pile 21	Bay 3
Group 6 (10000-16500)	1750	Pile 22	

Scenario 5:

Group (Length Group)	Pile Height	Pile	
Group 5 (3350-11000)	1750	Pile 1	Bay 1
Group 4 (4000-11000)	1750	Pile 2	
Group 5 (11000-16500)	1750	Pile 3	
Group 3 (10000-17000)	1750	Pile 4	
Group 4 (11000-16500)	1750	Pile 13	
Group 11 (9000-14000)	1750	Pile 14	
Group 10 (10000-17000)	1750	Pile 15	Bay 2
Group 1 (10000-15500)	1750	Pile 16	
Group 6 (4000-10000)	1750	Pile 5	
Group 8 (4000-9000)	1750	Pile 6	
Group2 (1000-16000)	1750	Pile 7	
Group 7 (10000-17000)	1750	Pile 8	
Group 8 (9000-15000)	1750	Pile 17	Bay 3
Group 10 (3000-10000)	1750	Pile 18	
Group2 (3400-10000)	1750	Pile 19	
Group 6 (10000-16500)	1750	Pile 20	
Group 7 (4000-10000)	1750	Pile 9	
Group 9 (9000-16500)	1750	Pile 10	
Group 3 (3100-10000)	1750	Pile 11	
Group 9 (3700-9000)	1750	Pile 12	
Group 11 (3000-9000)	1750	Pile 21	Bay 3
Group 1 (3400-10000)	1750	Pile 22	

Scenario 6:

Group (Length Group)	Pile Height	Pile	
Group 5 (3350-11000)	1000	Pile 1	Bay 1
Group 4 (4000-11000)	1000	Pile 2	
Group 5 (11000-16500)	1000	Pile 3	
Group 3 (10000-17000)	1000	Pile 4	
Group 4 (11000-16500)	1000	Pile 13	
Group 11 (9000-14000)	1000	Pile 14	
Group 10 (10000-17000)	1000	Pile 15	
Group 1 (10000-15500)	1000	Pile 16	
Group 6 (4000-10000)	1000	Pile 5	Bay 2
Group 8 (4000-9000)	1000	Pile 6	
Group2 (1000-16000)	1000	Pile 7	
Group 7 (10000-17000)	1000	Pile 8	
Group 8 (9000-15000)	1000	Pile 17	
Group 10 (3000-10000)	1000	Pile 18	
Group2 (3400-10000)	1000	Pile 19	
Group 6 (10000-16500)	1000	Pile 20	Bay 3
Group 7 (4000-10000)	1000	Pile 9	
Group 9 (9000-16500)	1000	Pile 10	
Group 3 (3100-10000)	1000	Pile 11	
Group 9 (3700-9000)	1000	Pile 12	
Group 11 (3000-9000)	1000	Pile 21	
Group 1 (3400-10000)	1000	Pile 22	

Scenario 7:

Group (Length Group)	Pile Height	Pile	
Group 5 (3350-11000)	1000	Pile 1	Bay 1
Group 4 (4000-11000)	1000	Pile 2	
Group 5 (11000-16500)	1000	Pile 3	
Group 3 (10000-17000)	1000	Pile 4	
Group 4 (11000-16500)	300	Pile 13	
Group 11 (9000-14000)	300	Pile 14	
Group 10 (10000-17000)	300	Pile 15	
Group 1 (10000-15500)	300	Pile 16	
Group 6 (4000-10000)	1000	Pile 5	Bay 2
Group 8 (4000-9000)	1000	Pile 6	
Group2 (1000-16000)	1000	Pile 7	
Group 7 (10000-17000)	1000	Pile 8	
Group 8 (9000-15000)	300	Pile 17	
Group 10 (3000-10000)	300	Pile 18	
Group2 (3400-10000)	300	Pile 19	
Group 6 (10000-16500)	300	Pile 20	Bay 3
Group 7 (4000-10000)	300	Pile 9	
Group 9 (9000-16500)	1000	Pile 10	
Group 3 (3100-10000)	300	Pile 11	
Group 9 (3700-9000)	300	Pile 12	
Group 11 (3000-9000)	300	Pile 21	
Group 1 (3400-10000)	300	Pile 22	

Appendix F: Scenario 2

Group (Length Group)	Pile Height	Pile	Bay #	Total Pile Height/Week	No of pile disposals/week
5 (3350-11000)	1750	Pile 1	Bay 1	12064.17	6.893811429
6 (4000-10000)	1750	Pile 2		11801.03	6.743445714
7 (4000-10000)	1750	Pile 3		9570.64	5.468937143
9 (9000-16500)	1750	Pile 4		7636.2	4.363542857
8 (4000-9000)	1750	Pile 13		6462.84	3.693051429
4 (4000-11000)	1750	Pile 14		5620.63	3.211788571
2 (10000-16000)	1750	Pile 15		4595.02	2.625725714
3 (3100-10000)	1750	Pile 16		4214.92	2.408525714
9 (3700-9000)	1750	Pile 5	Bay 2	3724.28	2.12816
6 (10000-16500)	1750	Pile 6		3447.76	1.970148571
10 (3000-10000)	1750	Pile 7		3293.19	1.881822857
2 (3400-10000)	1750	Pile 8		3233.75	1.847857143
8 (9000-15000)	1750	Pile 17		2861.13	1.634931429
5 (11000-16500)	1750	Pile 18		2800.75	1.600428571
7 (10000-17000)	1750	Pile 19		2607.01	1.48972
11 (3000-9000)	1750	Pile 9	Bay 3	2115.65	1.208942857
3 (10000-17000)	1750	Pile 10		1848.59	1.056337143
1 (10000-15500)	1750	Pile 11		1694.79	0.9684514286
4 (11000-16500)	1750	Pile 12		1659.71	0.9484057143
11 (9000-14000)	1750	Pile 21		1260.82	0.7204685714
1 (3400-10000)	1750	Pile 22		1250.62	0.71464
10(10000-17000)	1750	Pile 23		869.64	0.4969371429

